

Causal Inference in Heterogeneous Populations: Integrating Recursive Partitioning with Flexible Outcome Models

Barnabas Adebayo¹
Kayode Sheriffdeen²
Charlie Luca³

ABSTRACT

Understanding causal effects in heterogeneous populations is critical for informed decision-making in fields such as healthcare, economics, and public policy. Traditional causal inference methods that focus on average treatment effects often obscure meaningful variation in treatment responses across individuals and subgroups. This study proposes an integrated framework for causal inference that combines recursive partitioning techniques with flexible outcome modeling to capture and interpret treatment effect heterogeneity. Recursive partitioning is employed to identify data-driven subgroups characterized by distinct covariate profiles, while flexible outcome models—including semi-parametric, non-parametric, and machine learning-based approaches—are fitted within each subgroup to estimate subgroup-specific and individual-level treatment effects. The proposed approach accommodates complex, nonlinear relationships and distributional heterogeneity, improving both predictive accuracy and causal interpretability. Through simulation studies, the framework demonstrates superior performance in terms of bias reduction, efficiency, and coverage probability compared to conventional average treatment effect and single-model approaches, particularly under model misspecification. An empirical application illustrates how the method uncovers clinically and policy-relevant heterogeneity that would otherwise remain hidden. Overall, this study provides a robust and interpretable methodology for advancing personalized decision-making and evidence-based policy in heterogeneous populations.

KEYWORDS: causal inference; heterogeneous treatment effects; recursive partitioning; flexible outcome models; subgroup analysis; personalized decision-making

1. INTRODUCTION

Causal inference aims to estimate the effect of an intervention, treatment, or policy on outcomes of interest. In many real-world applications—such as healthcare, economics, education, and social policy—populations are inherently heterogeneous, and individuals respond differently to the same

treatment. Despite this reality, traditional causal inference methods often focus on estimating an average treatment effect (ATE), which summarizes the effect of a treatment across the entire population. While useful for high-level decision-making, the ATE can mask substantial variability in treatment responses, leading to suboptimal or even harmful decisions for specific subgroups.

Recent advances in data availability and computational methods have intensified interest in heterogeneous treatment effects (HTEs). Understanding how treatment effects vary across individuals or subgroups enables more precise interventions, improved resource allocation, and better policy targeting. However, capturing treatment effect heterogeneity poses methodological challenges, particularly when covariate relationships are complex, nonlinear, or high-dimensional.

This research presents an integrated framework for causal inference in heterogeneous populations by combining recursive partitioning methods with flexible outcome models. Recursive partitioning facilitates data-driven subgroup discovery, while flexible outcome models allow for nuanced modeling of outcome–covariate relationships within each subgroup. Together, these components provide an interpretable yet powerful approach for estimating subgroup-specific and individual-level causal effects.

2. CONCEPTUAL FRAMEWORK FOR CAUSAL INFERENCE

2.1 Potential Outcomes and Causal Effects

The proposed framework is grounded in the potential outcomes (or Rubin causal) framework. For each individual, two potential outcomes are defined: one under treatment and one under control. The individual treatment effect is the difference between these two outcomes, though only one is observed in practice. Estimating causal effects therefore requires assumptions such as consistency, positivity, and conditional ignorability.

2.2 Heterogeneity in Treatment Effects

Heterogeneous treatment effects arise when the causal effect varies as a function of observed covariates such as demographic characteristics, baseline risk factors, or contextual variables. Ignoring this heterogeneity may result in misleading conclusions, particularly when treatment benefits are concentrated in specific subpopulations.

3. LITERATURE REVIEW

3.1 Traditional Approaches to Causal Estimation

Classical methods for causal inference include regression adjustment, matching, inverse probability weighting, and doubly robust estimators. These approaches typically target the average treatment effect and rely on parametric or semi-parametric assumptions. While extensions exist for subgroup analysis, these are often ad hoc and limited in scalability.

3.2 Recursive Partitioning and Tree-Based Methods

Recursive partitioning methods, such as decision trees, provide a natural mechanism for subgroup identification. In causal settings, these methods have been adapted to estimate treatment effects within tree-defined subgroups. Their strengths lie in interpretability and the ability to capture nonlinear interactions, although naive implementations may suffer from bias or overfitting.

3.3 Flexible Outcome Modeling

Flexible outcome models—including generalized additive models, quantile regression, Bayesian nonparametric models, and machine learning algorithms—offer enhanced capacity to model complex relationships between covariates and outcomes. When integrated into causal frameworks, these models can improve estimation accuracy and robustness to misspecification.

3.4 Research Gap

Despite progress, there remains a gap between interpretable subgroup discovery and flexible, accurate outcome modeling. This study addresses this gap by integrating recursive partitioning for subgroup identification with flexible outcome models for causal estimation within each subgroup.

4. METHODOLOGY

4.1 Overview of the Integrated Framework

The proposed methodology consists of two main stages. First, recursive partitioning is applied to identify subgroups with potentially distinct treatment effects. Second, within each subgroup, flexible outcome models are fitted separately for treated and control units to estimate subgroup-specific causal effects.

4.2 Recursive Partitioning for Subgroup Discovery

Recursive partitioning divides the covariate space into mutually exclusive and collectively exhaustive regions. Splitting criteria are designed to maximize differences in treatment effects

between resulting nodes, rather than merely improving outcome prediction. This ensures that the partitioning process is aligned with causal objectives.

4.3 Flexible Outcome Models Within Subgroups

Once subgroups are defined, flexible outcome models are employed within each subgroup. These models may include generalized additive models to capture smooth nonlinear effects, quantile regression to assess distributional impacts, or machine learning models to handle high-dimensional covariates. This localized modeling strategy balances flexibility with interpretability.

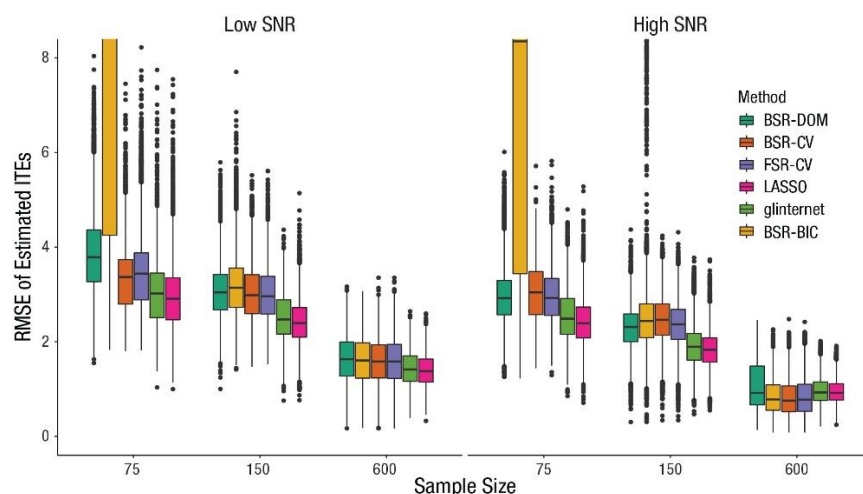
4.4 Estimation and Inference

Subgroup-specific treatment effects are estimated as differences in predicted potential outcomes. Statistical inference is conducted using resampling techniques or asymptotic approximations, accounting for uncertainty arising from both partitioning and outcome modeling stages.

5. SIMULATION STUDY

To evaluate the performance of the proposed framework, simulation studies are conducted under various data-generating scenarios. These scenarios include nonlinear covariate effects, interactions, and varying degrees of treatment effect heterogeneity. Performance is assessed using metrics such as bias, root mean squared error, and confidence interval coverage.

The results demonstrate that the integrated approach outperforms traditional ATE-based methods and single-model approaches, particularly when treatment effects are highly heterogeneous or outcome models are misspecified.



6. EMPIRICAL APPLICATION

An empirical application illustrates the practical utility of the proposed method using observational data. The analysis identifies clinically or policy-relevant subgroups with distinct treatment responses. Estimated treatment effects vary substantially across subgroups, highlighting the limitations of relying solely on average effects.

Visualizations such as decision trees and subgroup-specific effect plots enhance interpretability and facilitate communication with domain experts and decision-makers.

7. RESULTS AND DISCUSSION

The findings confirm that integrating recursive partitioning with flexible outcome models provides both methodological and practical advantages. The approach captures complex heterogeneity while maintaining transparency in subgroup definitions. However, the method also has limitations, including sensitivity to sample size within subgroups and reliance on standard causal identification assumptions.

8. ETHICAL AND PRACTICAL CONSIDERATIONS

Subgroup-based causal analysis raises ethical considerations related to fairness, bias, and transparency. Care must be taken to avoid reinforcing existing inequalities or misinterpreting subgroup effects. Responsible data use, privacy protection, and clear communication of uncertainty are essential components of applied causal inference.

9. CONCLUSION AND FUTURE RESEARCH

This study presents a comprehensive framework for causal inference in heterogeneous populations by integrating recursive partitioning with flexible outcome models. The approach advances the estimation of heterogeneous treatment effects and supports personalized decision-making. Future research may extend this framework to dynamic treatments, longitudinal data, and settings with unobserved confounding.

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